The law of excluded middle is not a theorem of type theory

The Univalent Foundations Program

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Editorial note. The counted statement is the author cor environment at source lines 234-237 and its proof environment opens at line 238, the next line; the pairing is by adjacency and the retained proof range 238-246 is the whole environment.

Editorial note. The retained counter values (chapter 3, section 2, theorem offset 6) make the counted corollary print as Corollary 3.2.7, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:not-lem = 3.2.7).

Editorial note. One label the proof cites lies outside the card, thm:not-dneg (book Theorem 3.2.2, defined at line 177 of the same pinned file), and is declared in external_references as a non-load-bearing navigation pointer: it is counted by its own card of this lane and the retained proof carries out the whole derivation of the implication it needs.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the law of excluded middle is not provable (author label thm:not-lem, book Corollary 3.2.7).

Allowed standard prerequisites (editorial).

- Law of excluded middle as a type. The law of excluded middle is the type of functions sending each type A to an element of the coproduct of A with its negation.. It is the hypothesis refuted.
- Failure of double negation. It is not the case that every doubly negated type implies the type itself.. It is the earlier result the argument is reduced to, and it is retained as a navigation pointer.
- Case analysis on a coproduct. An element of a coproduct may be assumed to be in either summand, and the empty type yields any conclusion.. It is how each of the two cases is discharged.

Source reference. Original theorem 3.2.2: sources/hott-logic/pinned-original-source.tex, source lines 177–177. Book Theorem 3.2.2, the failure of the law of double negation, cited at source line 241. Navigation pointer only: it is counted

by its own card of this lane and nothing of the counted proof is delegated to it beyond that one application.

Corollary 3.2.7. *It is not the case that for all $A : \mathcal{U}$ we have $A + (\neg A)$.*

Proof. Suppose we had $g : \prod_{(A:\mathcal{U})}(A+(\neg A))$. We will show that then $\prod_{(A:\mathcal{U})}(\neg\neg A \to A)$, so that we can apply original theorem 3.2.2. Thus, suppose $A : \mathcal{U}$ and $u : \neg\neg A$; we want to construct an element of A.

Now $g(A) : A+(\neg A)$, so by case analysis, we may assume either $g(A) \equiv \mathsf{inl}(a)$ for some $a : A$, or $g(A) \equiv \mathsf{inr}(w)$ for some $w : \neg A$. In the first case, we have $a : A$, while in the second case we have $u(w) : \mathbf{0}$ and so we can obtain anything we wish (such as A). Thus, in both cases we have an element of A, as desired. $\square$